End-of-Project Reflective Report

Applied Industry Project for the Technology Investment Network (TIN200)

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Reflective framework: Gibbs' Reflective Cycle (1988)

1. Background

Technology Investment Network (TIN) is the New Zealand firm behind the TIN200, the country's main annual ranking of its largest technology exporters. Our five-person team was brought in through BUSINFO 718 to strengthen the evidence base behind the 2026 edition of that ranking. That meant a few distinct strands of work. We checked each candidate against TIN's inclusion criteria and built supporting datasets covering cleantech, early-stage investment, export context, acquisitions, and contacts. All of it then fed into a single, fully sourced data foundation, with automation handling the refresh. At every stage, the public evidence was there to support TIN's expert judgement rather than replace it. We handed over four deliverables: a consolidated research workbook, an economic analysis, an automation and reporting manual, and a client presentation.

2. Reflective Framework

I have structured this report around Gibbs' (1988) Reflective Cycle. It runs through six stages, from description and feelings to evaluation, analysis, conclusion, and action plan (see Figure 2.1). I chose Gibbs because the assessment asks for more than a description of the project. It wants me to evaluate the outcomes and draw out what they mean for my professional practice, and Gibbs' final two stages are built for exactly that. The model's attention to feelings matters too, because working in a team under pressure carried a real emotional charge that shaped how I reacted. The cycle does get criticised for nudging reflection into a straight line rather than a genuine loop (Dyment & O'Connell, 2011), so I have treated it as a recurring cycle in the spirit of experiential learning theory (Kolb, 1984), and used it to organise each of the three learning points below.

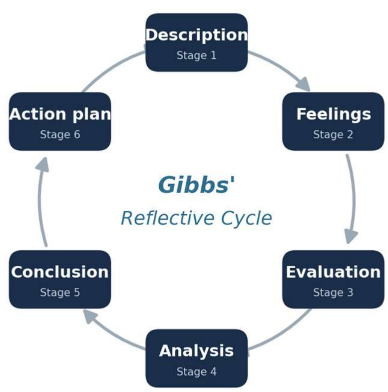

Figure 2.1
Gibbs' Reflective Cycle (1988)

Note. Adapted from Gibbs (1988). Each of the three learning points in Section 4 is organised through these six stages.

3. Project Evaluation

The project largely hit its objectives (see Table 3.1). The core deliverable was a validated working list: 145 companies retained, around 50 flagged for review, and five recommended for removal. The supporting datasets included a directory of 111 cleantech companies, a 2025 early-stage investment dataset (NZ$754m across 166 deals), sourced export figures placing technology as New Zealand's third-largest export earner at NZ$15.31b, and validation of about thirty acquisitions. All of it was held in a single, fully sourced workbook with a one-click refresh tool. The final presentation was well received, which I took as a sign we had met expectations. It was not a flawless result, though. The cleantech and emerging-company lists in particular were still watchlists that needed further validation, and we had deliberately left a batch of company-name matches to be resolved by hand. The foundation is solid, but there is more to do before it can be considered finished.

Table 3.1

The extent to which the project objectives were achieved

Objective	Outcome delivered	Assessment
Validate the 2026 candidate list against TIN's five criteria	200 companies reviewed: 145 retained, ~50 flagged for review, 5 removed; foreign ownership flagged for annual check	**Achieved**
Build supporting research datasets (cleantech, investment, exports, acquisitions, contacts)	111-company cleantech directory; 2025 investment (NZ$754m, 166 deals); export ranking; ~30 acquisitions validated; contacts recovered	**Largely achieved**
Deliver a consolidated, sourced and maintainable data foundation with automation	13-section sourced workbook, one-click refresh tool, and plain-English manual	**Achieved**
Support, not replace, TIN's expert judgement	Public evidence used only to support the confidential survey and final ranking decision	**Achieved by design**

The value to TIN is practical and lasting. We turned a largely manual, once-a-year exercise into an evidence base that is defensible, auditable, and repeatable. The TIN200 feeds investors and the government, so the credibility of the data underneath it really does matter, and a master list with recorded decisions and provenance cuts the risk of error in future editions.

The methodology had clear strengths and limitations (see Table 3.2).

Table 3.2

Strengths and limitations of the methodology applied

Methodological strengths	Methodological limitations
• Evidence graded by source strength, with official records (NZX, Companies Office, and Takeovers Panel) prioritised over media • Every figure sourced and dated; data anomalies flagged proactively • Company names standardised to prevent double-counting; findings triangulated	• Contact and ownership data are point-in-time and decay over time • Public evidence cannot replace TIN's confidential survey; pipeline presence does not confirm eligibility • Not every acquisition could be traced to equal depth; some relationships are associative, not causal

The approach held up well overall, even though it relied on public information that shifts over time and is never quite complete. That limitation surfaced twice in practice. Partway through, our headline investment figure had to be revised from NZ$467m to NZ$754m once fuller data came in, and while validating acquisitions I found one that had first been attributed to the wrong acquirer. Catching both came down to keeping close track of where each figure came from.

4. Learning Points

I take three main learning points from the project: one technical, one social, and one about data governance. Each is organised through the six stages of Gibbs' cycle, summarised in Table 4.1 and then discussed in turn.

Table 4.1

Summary of the three learning points

Learning area	Key experience	Core lesson	Future professional practice
Technical (4.1)	A revised investment figure, a corrected misattribution, and a "one-story" analysis of size metrics	Rigour means provenance, traceability, and restraint	Embed source-and-date metadata from the outset; favour parsimonious, well-justified measures
Social (4.2)	A team-conduct issue and a unilateral client contact that strained the relationship	Set team norms early; separate behaviour from character in conflict	Agree on client-contact and decision protocols early; address friction promptly; keep escalation factual
Governance (4.3)	Insisting on provenance and applying FAIR and CARE thinking to the data	Data stewardship is an ethical, not merely technical, practice	Apply FAIR and CARE to how data is sourced, attributed, and reused; weigh context

4.1 Technical: data integrity, provenance and analytical restraint

Description and feelings. I led much of the project's data work: building the consolidated workbook in Python, validating acquisitions, and analysing the dataset. Two moments stand out. First, our headline early-stage investment figure moved from NZ$467m to NZ$754m once full-year 2025 data came in. That made me uneasy at the time, because a number that changes can look like a mistake to a client. Second, while validating acquisitions, I found that one company had been pinned on the wrong acquirer in an earlier draft, and I corrected it before it reached TIN. That one was quietly satisfying. I also stepped back to check whether our many measures of company size were actually independent, and it turned out they more or less told the same story.

Evaluation and analysis. Sourcing and dating every figure was the discipline that paid off. Because the provenance was on record, the revised investment number held up as a matter of currency rather than error, and the misattribution was easy to trace and fix. What I would do better is build "verified-on" metadata into the dataset from the start instead of retrofitting it. This all comes back to the fact that public data is provisional by nature. Data quality is multidimensional, taking in timeliness and credibility as well as accuracy (Wang & Strong, 1996), so a figure that moves as better data arrives reflects currency, not failure. Rigour in a data project is as much about traceability and restraint as it is about computation, and reporting a few well-justified measures proved more useful than piling up many overlapping ones.

Conclusion and action plan. What I took from this is that technical credibility rests on humility and provenance, not on the sheer volume of analysis. Going forward, I will build source-and-date metadata into datasets from the start, flag anomalies early and openly, and favour a few well-reasoned measures over a long list of redundant ones.

4.2 Social: navigating team conflict and client communication

Description and feelings. Midway through the project, friction built up in the team over process and conduct, including one instance where someone contacted the client on their own in a way that strained the relationship with TIN. Along with two colleagues, I took our concerns to the course coordinator formally, looking for a factual record, some mediation, and a way to repair the client relationship rather than to hand out blame. I found it genuinely stressful. I had prior professional experience to draw on, but the peer dynamic of a student team was new to me. I was very aware of fairness, and of protecting both the client relationship and my teammates.

Evaluation and analysis. Looking back, the framing was the part I got right. I kept the focus on specific incidents rather than personalities, aimed for a constructive outcome, and the client relationship came through intact. If I were doing it again, I would act sooner. I should have raised the process concerns directly and informally while they were still small, and pushed early for a clear agreement on who was allowed to contact the client. It came down to never having agreed clear norms around decision rights or client contact, and that gap left room for someone to act alone. My years in industry had taught me to fall back on documentation and process, which gave the response structure and a sense of fairness. The cost was that the same reflex sent me to formal channels when a quieter, earlier conversation might have settled things before they reached the client.

Conclusion and action plan. The main takeaway was the importance of agreeing on communication protocols and decision rights early, while a team is still forming, and of treating conflict as being about behaviour rather than character. In future work, I will put client-contact and decision-making protocols in place from the start, raise friction quietly and without delay, and keep any escalation factual, proportionate, and aimed at resolving the problem.

4.3 Data governance and ethical data stewardship

Description and feelings. Throughout the project, I treated the TIN200 data as something investors and policymakers actually rely on, holding the line on provenance and refusing to “fill gaps” with unverified figures. My coursework on data governance, on the FAIR and CARE principles, and on Māori Data Sovereignty shaped how I thought about responsibility for data. I felt a real sense of stewardship, and now and then a tug-of-war between speed and integrity whenever a quick, unsourced answer was tempting.

Evaluation and analysis. I kept the sourcing consistent throughout, and nothing unverified made its way into the client deliverables. Where I fell short was on the part of governance that goes past accuracy: who can access the data, how it is attributed, and whose interests end up embedded in it. Working through these frameworks, I realised that even though the TIN data is commercial rather than Indigenous, the underlying ethics still apply. The FAIR principles of findable, accessible, interoperable, and reusable data (Wilkinson et al., 2016) map onto how I structured and documented the workbook. The CARE principles of collective benefit, authority, responsibility, and ethics (Carroll et al., 2020) pushed me further, reframing accuracy as a duty to the people represented in and affected by the data, not just a quality metric. That shift happened because my operations background had trained me to chase accuracy, while these frameworks taught me to treat data as carrying obligations.

Conclusion and action plan. I came away seeing data stewardship as an ethical practice, not just a technical one. Going forward, I will bring FAIR and CARE thinking to how data is sourced, attributed, and reused, and I will weigh context deliberately, because different data carries different obligations and good governance is part of being good at the job.

5. Conclusion

The project met its objectives and gave TIN a credible, reusable, and auditable evidence base in place of what used to be a manual annual exercise. For a client whose work feeds investment and policy decisions, that is a genuinely useful result. What mattered more to me, though, was what the project built in me (see Table 4.1). On the technical side, rigour came to mean provenance and restraint as much as the analysis itself. Socially, I got a much clearer sense of how to set team norms and work through conflict constructively. Ethically, I developed a more mature view of data as something held in trust. Bridging more than a decade of industry experience with academic frameworks turned out to be surprisingly powerful. The frameworks gave language to instincts I had built up over years on the job, and the project let me test them against a live client and team rather than a case study. I will carry a few of those lessons into analytics and consulting. Rigour still matters to me, but I have learned to pair it with the humility to know when I am wrong. I would raise friction with colleagues earlier and more openly than I did here. And I no longer treat the ethics of data as separate from the technical side of the work. Academically, the project showed me where my analytical foundations need deepening, and I will carry that into how I approach further study.

6. References

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7. AI Use Statement

Generative AI tools were used in a supporting role to help structure and refine the writing of this report. All reflections, analysis and conclusions are my own, based on my involvement in the project.